

PIYUSH KUMAR

Data Scientist · Supply Chain Analytics · ML | MLOps | GenAI · 4+ Years
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Data Scientist with 4+ years delivering supply chain analytics across manufacturing and e-commerce - specialising in **time series forecasting, optimisation, simulation, and supplier risk modelling** aligned to S&OP, procurement, and inventory decisions. Proficient in Python, PySpark, SQL, Oracle OCI Data Science, AWS, Azure, and Power BI; demonstrated impact: **₹1 Cr+ savings, 18% inventory reduction, and 12% margin gains.**

SKILLS

ML & Forecasting: Python, PySpark, XGBoost, LightGBM, Prophet, ARIMA/SARIMA, Hierarchical Forecasting, Scikit-learn, SHAP

Optimisation & Simulation: Linear/Mixed-Integer Programming (PuLP/SciPy), EOQ, Safety Stock, Monte Carlo, SimPy, RL-based Pricing

MLOps & Cloud: Oracle OCI Data Science, AWS (SageMaker/S3/Lambda), Azure ML, MLflow, Docker, GitHub Actions

Data & BI: SQL (Window Functions, CTEs), Power BI, Tableau, ERP/IoT Pipelines, REST APIs

SC Domain & Tools: S&OP, Demand Forecasting, Procurement Analytics, Supplier Risk, Inventory & Network Planning · Awareness: SAP IBP, o9, Kinaxis

EXPERIENCE

Data Scientist — Jayanita Exports (Manufacturing & Export)

Jan 2025 – Present

- Built demand forecasting system (Prophet + XGBoost, 500+ SKUs) achieving 15% MAPE improvement; outputs feed S&OP demand review, reducing stockout risk across ₹20 Cr+ procurement decisions.
- Designed procurement analytics pipeline — XGBoost price forecasting, EOQ-based safety stock optimisation, SHAP-explained supplier risk scoring; delivered **₹1 Cr+ annual savings** and 8% raw-material cost reduction.
- Built inventory optimisation models (ABC analysis, reorder-point, SKU profitability scoring): cut holding costs **18%**, lifted contribution margin **12%**, drove **10% YoY revenue growth**.
- Automated KRA/KPI reporting (ML + Power BI dashboards); eliminated 1,200+ man-hours/year, reduced unit costs 7%; deployed ERP + IoT anomaly detection pipeline improving defect detection 20%.

Software Developer & Data Analyst — Laysan Technologies Pvt. Ltd.

Sep 2021 – Dec 2024

- Led 4–5 person team building Taste2Plate (MERN platform) with Python + Power BI analytics layer; built REST APIs and automated ETL pipelines enabling clients to scale platforms **35% faster**.

PROJECTS

Integrated Demand-Supply Intelligence Platform — Procurement, Forecasting & Optimisation Engine · Python, SQL, XGBoost, Prophet, ARIMA, SHAP, PuLP, AWS, Power BI

- Built ERP + BOM + supplier + commodity/FX data pipeline using **6-8 table SQL joins**; benchmarked ARIMA, Prophet and XGBoost with walk-forward CV, achieving **4.8% MAPE** for 4-12 week raw-material price and demand forecasts.
- Designed supplier risk scoring plus multi-echelon procurement optimisation in PuLP with MOQ, lead-time, storage and ABC service-level constraints; delivered **₹1 Cr+ savings, 8% RM cost reduction and 18% lower inventory holding**.

Network-Level S&OP Decision Engine — Personal Benchmark / OCI Prototype · Python, Prophet, LightGBM, PuLP, SimPy, OCI Data Science, Streamlit

- Prototyped hierarchical demand forecasting across **SKU -> family -> network** levels with reconciliation logic for a 3-plant, 5-DC, 50-retail-node supply chain, supporting S&OP demand-supply rebalancing.
- Implemented MILP production/network planning and SimPy discrete-event simulation to stress-test stochastic demand and lead times; generated targeted safety-stock recommendations and an **Oracle OCI Data Science-ready** deployment pattern.

EDUCATION

MBA, Business Analytics — Sharda University · 82.43%

2023 – 2025

Coursework: Time Series Forecasting · Operations Research & Optimisation · Supply Chain Management

B.Tech, Computer Science & Engineering — AKTU University · 77.34%

2019 – 2023

ACHIEVEMENTS & CERTIFICATIONS

- 24+ national data science competitions — Top-5 finishes · DecodeX 2024 (Top 4): XGBoost + SHAP on 650-variable dataset · Renaissance 2024 (3rd): predictive optimisation, 5,000+ residents
- Oracle Cloud Infrastructure Foundations (OCI) · AWS Fundamentals · BCG Data Science · PwC Power BI · Accenture & TATA Data Visualisation